Nguyễn Ngọc Huy

Da Nang, Viet Nam

0355621838 huynguyenngoc.dev@gmail.com huynguyenngocdev.github.io

OBJECTIVE

I am a programmer driven by a relentless passion for coding and an insatiable hunger for learning. I believe that the best products are built at the intersection of deep dive technical skill and a 'never-stop-learning' mindset. For me, technology isn't just a career—it's a journey of continuous growth where I embrace new challenges and emerging tools like AI with open arms.I possess a powerful ability to self-teach and rapidly integrate new technologies into my workflow, whether it's mastering a new framework or leveraging AI to accelerate development. With over 3 years of experience, I've specialized in crafting high-impact solutions for Smart Home, IoT, and E-commerce environments. I pride myself on being a strong, independent thinker who can delve deeply into complex problems alone; yet, I thrive just as much in Scrum/Agile teams, leading sprints and guiding projects. My ultimate goal is simple: to transform cutting-edge code into real-world value that makes life easier, safer, and more enjoyable for every user.

EDUCATION

PASSERELLES NUMÉRIQUES VIETNAM-DANAVTC - *Major: Information Technology* Sep 2019 - Oct 2022

- - GPA: 3.8/4

WORK EXPERIENCE

Digital Fortress - *Mobile Developer* Sep 2022 - Present

- Designed and shipped multi-platform IoT applications (iOS, macOS, Catalyst) for Matter-compliant devices using SwiftUI, Combine, and Objective-C.Implemented device commissioning (QR/Manual/Thread/Wi-Fi), multi-admin onboarding, OTA updates, and cluster-based controls with real-time attribute syncing and local caching.
- Engineered a networking layer using NetworkExtension, Bonjour/mDNS, and NWBrowser to discover Thread networks and persist OTBR credentials.
- Architected large-scale Flutter applications following Clean Architecture and MVVM, featuring WebRTC video streaming, MQTT event control, and BLE/QR provisioning.
- Developed an iOS palm-biometrics security app using ONNX Runtime, MediaPipe, and TensorFlow Lite for feature extraction and liveness detection.Integrated on-device LLM inference to analyze health data from Apple Health and Google Fit for personalized user warnings.
- Implemented end-to-end security with TLS pinning (TOFU), App Attest/DeviceCheck, anti-jailbreak checks, and secure storage using CryptoKit.
- Built modular Swift Packages with Objective-C bridges for CoAP and DTLS, and integrated Tuya SmartLife SDK via MethodChannels and Kotlin.
- Managed subscription flows with StoreKit 2 and localized applications into six languages (EN, DE, ES, FR, IT, ZH-Hans).Upgraded React Native projects (v0.74 to v0.80) and integrated third-party SDKs, iPay88, and Firebase.Streamlined release cycles by 40% through CI/CD setup (GitHub Actions).
- Acted as Technical Lead and Scrum Master, facilitating sprint planning, conducting code reviews, mentoring interns and freshers, and conducting workshops on Flutter and SwiftUI.Collaborated in an Agile environment with overseas clients and cross-functional teams, demonstrating English communication and problem-solving skills under pressure.

Umbala Metaverse - *Front-end & Mobile Developer (Intern)* Apr 2022 - Aug 2022

- Built the mobile version of the company's landing page using Flutter, ensuring responsive design and cross-platform consistency.
- Contributed to an internal Flutter app, implementing features such as MetaMask login (Web3 wallet authentication) and improving API integration using Retrofit.
- Practised SOLID, MVC/MVP/MVVM patterns and Clean Architecture, and strengthened skills in state management with Provider and BLoC.
- Wrote unit tests, integrated dependency injection and deep linking, and enhanced code quality through refactoring and design pattern application.

Code Engine Studio - *Back-end Developer (Intern)* Apr 2022 - Aug 2022

- Developed a RESTful API using ExpressJS for a Google Classroom-like platform; designed relational and NoSQL schemas and implemented authentication (username/password, Google OAuth) and file uploads to Google Drive.
- Collaborated with the team to design database workflows, apply API principles and implement secure token-based authentication using JWT.
- Practised SQL, MongoDB, OAuth, and API documentation techniques while improving problem-solving and collaboration skills.

CERTIFICATIONS

- HakkerRank Problem Solving 5 stars (hackerrank.com/huynguyenngoc_d1) 2022
- Flutter Bootcamp Certification
- Completed Internship at Umbala Metaverse

Completed Internship at Code Engine Studio 2021

HONORS & AWARDS

- Top 1 in the "Moment of Trust" contest in Passerelles numériques Vietnam 2022
- Little Star Project in the top 100 (https://startupwheel.vn/vi/cong-bo-top-100-startup-wheel-2022-vietnam/)

"Young Talent" Award at Digital Fortress Company 2023

"Sao vàng cống hiến" Award at Digital Fortress Company 2024

SKILLS

Technical skills	Flutter, SwiftUI, UIKit, React Native, TypeScript, ExpressJS, HTML/CSS
Soft skills	Agile&Scrum, leadership, mentoring, project estimation, requirements analysis, backlog grooming, risk management, effective communication (English/Vietnamese)

PROJECTS

1. Matter Utilities (SwiftUI)
Cross-platform smart-home application for iOS/macOS/Catalyst enabling commissioning and control of Matter devices. Implemented complex networking flows for Wi-Fi and Thread, multi-admin onboarding, cluster-level control, network discovery and OTBR management. Built an automation testing engine, generating test scenarios, executing sequential runs and exporting JSON reports. Optimised stability through attribute subscriptions and local caching, synchronising UI in real time. Integrated in-app logs, diagnostic uploads (S3), feedback pipeline (screenshots + Slack), StoreKit 2 subscriptions and multi-language support.

2. Smart Camera & Alert System (Flutter)
Large-scale Flutter app employing Clean Architecture and modularisation and released in production on both the Play Store and App Store. Enabled hub onboarding via BLE, Wi-Fi scanning and credential provisioning; integrated multiple IoT devices (RTSP camera, doorbell, garage door, alarm, sensors). Implemented real-time video streaming with WebRTC and event control via MQTT; built comprehensive authentication (email/password, OTP, social login), push notifications (FCM) and deep linking. Developed AI features such as voice assistant and face/vehicle collection management; designed camera zone editor for polygon-based surveillance; added Live Activity/Dynamic Island support on iOS. Established robust CI/CD with multiple flavors and GitHub Actions.

3. NFC Business Card (Flutter)
Digital business-card app supporting NFC and QR exchanges and deployed to both the Play Store and App Store. Built modular architecture with BLoC state management; implemented registration/login (social auth via Google/Facebook/Apple), profile management, NFC device activation and contact sharing via deep links. Added analytics dashboards (views, taps, connection rate) and push notifications. Provided multi-language support and secure storage (Hive, flutter_secure_storage).

4. Tuya SmartLife Integration (Flutter & Android Native)
Tuya SmartLife Integration (Flutter & Android Native)Integrated Tuya/ThingClips SDK into Flutter through MethodChannels and Kotlin bridging. Implemented user management, home creation, device pairing (BLE scan, EZ Wi-Fi pairing), device info retrieval and control for Tuya smart lights. Managed device removal and state publishing via Kotlin channels and exposed Flutter APIs. Configured Gradle dependencies and Manifest permissions; handled BLE/location permissions on Android.

5. Palm Biometrics Security (iOS)
Developed a secure authentication app using ONNX Runtime, MediaPipe, TensorFlow Lite and Vision/CoreML for palm segmentation, feature extraction and hybrid face/palm liveness detection. Implemented registration and authentication flows, gesture-based unlocking, multi-environment configuration (dev/staging/prod), full settings and transfer flows. Wrote parity tests to ensure model consistency across iOS and Android; enforced code quality with SwiftLint and continuous integration via Travis CI. Added analytics, crash monitoring and secure networking with Alamofire and RSSwiftNetworking.

6. Smart Plug & Smart Bulb Control (iOS & SwiftUI)
Designed modular monorepo architecture using Swift Package Manager (AppCore, AppState, Networking, Data, MatterHelper). Implemented real-time device control via REST and WebSocket (NWConnection), BLE/QR onboarding, grouping and scene management, and persisted data via CoreData. Added local and remote notifications, device provisioning via ESPProvision.

7. NFT Photo Frame (React Native)
React Native 0.74 application released on the Play Store and App Store that allows users to manage and display NFT artwork on connected photo frames. Built navigation using Native Stack and bottom tabs, NativeWind for Tailwind-style styling, and state management via Context API and reducer patterns. Implemented API communication with Axios interceptors and automatic token refresh; added real-time features using WebSocket and NetInfo for network monitoring and reconnection. Integrated Web3 libraries (WalletConnect/AppKit, Wagmi)

INTERESTS

Music, Anime, Gundam and Machine Keyboard